Lecture 3: Fracture Mechanics and Damage Tolerance

UCK 364E Design of Machine Elements

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Today

1. Why the previous lecture is not enough
2. Why ordinary stress analysis fails at a crack
3. The stress intensity factor $K_{\mathrm{I}} = \beta\sigma\sqrt{\pi a}$
4. Fracture toughness, constraint, and the strength–toughness trade
5. Fatigue crack growth and the Paris law
6. Residual strength and the critical crack
7. The inspection interval — where it all comes together
8. Safe-life, fail-safe, damage tolerance
9. Three worked examples

Why the previous lecture is not enough

Last time we predicted *initiation*

S–N for long lives, ε–N for notches. Both assume the material starts out sound and stop at the moment a crack appears.

- Real structure contains flaws from the start — inclusions, scratches, burrs at drilled holes, corrosion pits, the torn edge of a punched rivet hole. Initiation may be absent entirely
- Initiation scatter is enormous — a factor of ten or more. Crack *growth* is far more repeatable, so predictions built on it are more trustworthy
- A crack is survivable if you find it — the whole modern philosophy

The question changes

Lecture 2: *how long until a crack appears?*
Lecture 3: *given that a crack exists, how long until it is dangerous — and how often must I look?*

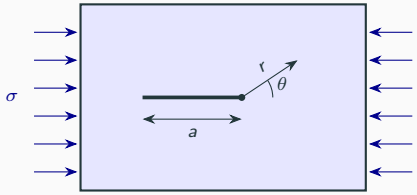
Why stress analysis fails at a crack

An ellipse flattened into a crack

$$K_t = 1 + 2\frac{a}{b} = 1 + 2\sqrt{\frac{a}{\rho}}$$

- Circular hole ($a = b$): the familiar $K_t = 3$
- As $\rho \to 0$: $K_t \to \infty$
- Elasticity says the stress at a sharp crack tip is infinite *whatever* the load

But cracked structures carry load, and identical cracks fail at the same stress. Something other than the peak stress controls fracture.



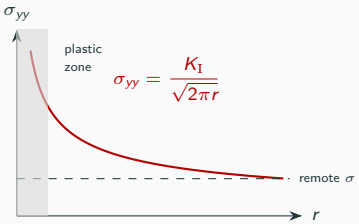
a crack in a plate

Irwin's resolution: characterise the *field*, not the peak

Near the tip of a crack in a linear elastic body,

$$\sigma_{ij}(r,\theta) = \frac{K}{\sqrt{2\pi r}}\, f_{ij}(\theta) + \dots$$

- The $r^{-1/2}$ singularity and the angular functions $f_{ij}(\theta)$ are the same for every crack in every body
- All of the geometry and loading sits in the single scalar K



the crack-tip field

Linear elastic fracture mechanics, in one sentence

Two different cracked components with the same K have the same crack-tip conditions and behave the same way.

Three modes: I opening, II sliding, III tearing. Mode I dominates — a crack under mixed loading turns normal to the maximum principal stress, converting itself into Mode I.

The stress intensity factor

Configuration	β	Note
Central crack $2a$, infinite plate	1.00	the reference case
Edge crack a, semi-infinite plate	1.12	the free edge relieves constraint
Two cracks at a hole, $a \gg R$	≈ 1.0	a measured from the hole centre
Short crack at a hole, $a \ll R$	≈ 3.4	the crack lies in the K_t field of the hole
Embedded penny crack	$2/\pi = 0.64$	fully constrained
Surface semi-elliptical crack	0.7–1.1	depends on a/c and a/t
Central crack, plate of width W	$\sqrt{\sec(\pi a/W)}$	finite-width correction

Three things students get wrong

1. $K \neq K_t$. K_t is dimensionless and describes a *notch*; K_I is in MPa$\sqrt{\text{m}}$ and describes a *crack*.
2. σ is the stress that would exist *without* the crack — not a local stress.
3. $K \propto \sqrt{a}$: doubling the crack raises K by only 41%. That is why growth is slow at first and catastrophic at the end.

The plastic zone — and when LEFM is allowed

$$r_p = \frac{1}{2\pi}\left(\frac{K_\mathrm{I}}{\sigma_y}\right)^2 \text{ (plane stress)}, \qquad r_p = \frac{1}{6\pi}\left(\frac{K_\mathrm{I}}{\sigma_y}\right)^2 \text{ (plane strain)}$$

- Infinite stress is impossible, so a small zone yields at the tip
- LEFM is valid only while that zone is small — small-scale yielding
- Rule of thumb: crack, ligament and thickness each $> 2.5(K/\sigma_y)^2$

When it is not valid

Elastic–plastic methods: J-integral, CTOD, or an R-curve. Thin aluminium fuselage skin is usually in this regime — which is why residual strength there uses crack-tip opening angle, not a single toughness number.

Fracture toughness

K_{Ic} is the plane-strain fracture toughness, measured to ASTM E399 on a specimen thick enough to guarantee plane strain:

$$B,\ a,\ (W-a)\ \geq\ 2.5\left(\frac{K_{\mathrm{Ic}}}{\sigma_y}\right)^2$$

Constraint

- **Thin**: free to contract through the thickness $\Rightarrow$ plane stress, big plastic zone, *high* apparent K_c
- **Thick**: contraction prevented $\Rightarrow$ plane strain, small zone, toughness falls to its *minimum* K_{Ic}
- Beyond the transition thickness it stops falling

Design with K_{Ic} unless you can justify the thin-section value by test.

Toughness and the critical crack

Material	σ_y (MPa)	K_{Ic}	a_c (mm)*	Comment
2024-T3 sheet	289	37	46.3	damage tolerant; fuselage skin
7075-T6	462	24	19.5	high strength, low toughness
7075-T73	400	31	32.5	overaged: toughness bought with strength
7050-T7451 plate	427	33	36.8	thick fittings
Ti-6Al-4V (ann.)	827	75	190.1	excellent combination
4340 (1800 MPa)	1517	50	84.5	modest toughness
300M (1900 MPa)	1600	65	142.8	landing gear — toughness is why

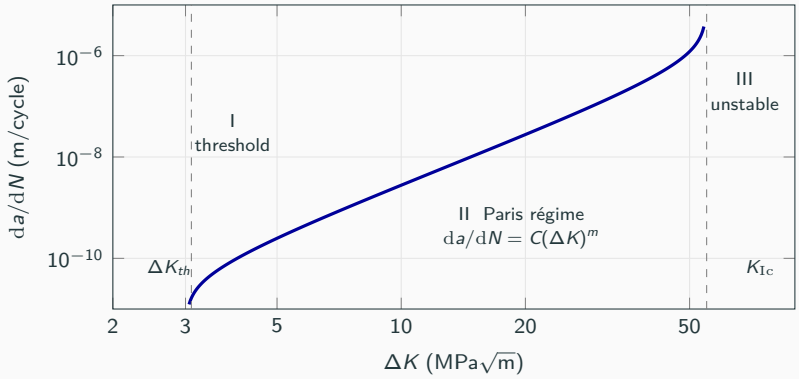
* a_c for a central crack ($\beta = 1$) at $\sigma = 0.5\sigma_y$. K_{Ic} in MPa $\sqrt{\mathrm{m}}$. Classroom values only.

The strength–toughness trade, quantified

7075-T6 is 60% stronger than 2024-T3 — and its critical crack is less than half as long. That is exactly why the *lower* wing surface and the fuselage are 2024, and the *upper* wing surface is 7075. Not "which alloy is better" but "which failure mode governs here".

Fatigue crack growth

The $\mathrm{d}a/\mathrm{d}N$ curve



I — threshold
below ΔK_{th} cracks do not propagate

II — Paris
straight on log–log; most of the life; where you calculate

III — unstable
$K_{\max} \to K_{\mathrm{Ic}}$, growth runs away

$$\frac{\mathrm{d}a}{\mathrm{d}N} = C(\Delta K)^m, \qquad m \approx 2.5\text{–}4$$

Because $\Delta K \propto \sqrt{a}$, the rate goes as $a^{m/2}$: at $m = 4$, a crack ten times longer grows a hundred times faster.

$$N_f = \frac{2}{(m-2)\, C \beta^m \Delta\sigma^m \pi^{m/2}} \left[\frac{1}{a_i^{(m-2)/2}} - \frac{1}{a_f^{(m-2)/2}} \right]$$

- Life is dominated by a_i. Halving the initial flaw beats doubling the final one
- The final crack size barely matters — at $m = 4$ you can often take $a_f \to \infty$
- Life scales as $\Delta\sigma^{-m}$ — a 10% stress reduction buys ≈46% more life at $m = 4$

In practice β varies with a: integrate numerically, cycle by cycle, with the Lecture 2 spectrum. That is what NASGRO and AFGROW do.

Mean stress, closure, retardation

R-dependence

- At higher mean stress the crack is open for more of the cycle and grows faster
- Forman and NASGRO equations add this explicitly

Crack closure

- The plastic wake behind the tip props the crack open
- Part of the nominal ΔK does no damage; only ΔK_{eff} drives growth

Retardation

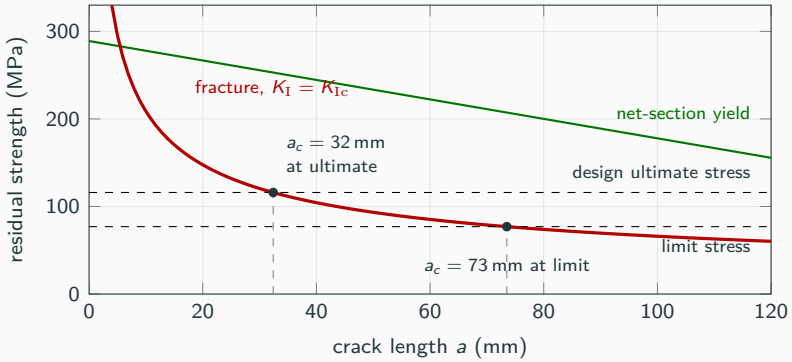
- One large overload makes an oversized plastic zone the crack must grow through
- Growth slows or stops for thousands of cycles

But it is ignored

Retardation is real and sometimes worth tens of percent of life — yet it is normally not credited in certification: relying on it is non-conservative, and a high–low sequence can equally produce *acceleration*.

Residual strength

The residual strength diagram



2024-T3, $K_{Ic} = 37\,\text{MPa}\sqrt{\text{m}}$, $\sigma_y = 289\,\text{MPa}$, central crack. Net-section yield governs when the crack is short; fracture governs when it is long.

What “critical” means

There is no single critical crack length

There is a crack length that is critical at a stated stress.

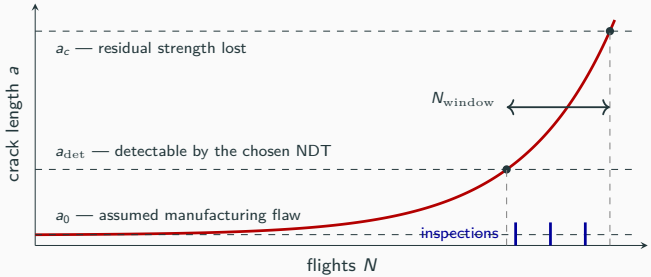
- CS-25.571 requires the *cracked* structure to carry a specified residual-strength load — commonly limit load, or a defined fraction, depending on the structural category and the inspection programme
- Choose the stress first, then read a_c off the diagram
- From the figure: $a_c = 32\,\text{mm}$ at ultimate stress, $73\,\text{mm}$ at limit stress — same material, same panel

The critical crack size is a *certification decision*, not a material property.

The inspection interval

Six steps

1. Assume an initial flaw a_0 — the largest that could escape manufacturing inspection; conventionally 1.27 mm (0.05 in) at a fastener hole
2. Compute a_c from the residual-strength requirement
3. Integrate the growth law over the flight spectrum: the curve $a(N)$ from a_0 to a_c, total life N_c
4. Determine the detectable size a_{det} for the method you will specify
5. The usable window runs from a_{det} to a_c: N_{window} flights
6. Repeat interval $= N_{\mathrm{window}}/n$, with $n \geq 2$ — so at least two inspections occur while the crack is detectable but not yet critical



Detectability is a design parameter

Method	Typical a_{det}	Notes
General visual, in service	25 mm to 50 mm	unaided, walkaround distance
Detailed visual, close	5 mm to 15 mm	clean surface, good light, unloaded
Dye penetrant	2 mm to 5 mm	surface-breaking only, surface accessible
Eddy current, surface	1 mm to 2 mm	finds cracks under rivet heads — the Aloha method
Ultrasonic	1 mm to 3 mm	subsurface; couplant, trained operator
X-ray	≈2% of thickness	poor for tight cracks normal to the beam

Formally detection is probabilistic: the probability of detection curve, and the certification figure $a_{90/95}$ — detected with 90% probability at 95% confidence.

The designer's lever is *access*

A crack that can only be found by removing an engine will be inspected rarely, the interval will be driven somewhere uneconomic — or the part must be safe-life.

Why this is a design lecture, not a materials lecture

Every item on the inspection-interval list is something you control:

- the initial flaw size — hole quality, cold expansion, finish
- the stress range — sizing, load path
- the critical crack size — material, tear straps, crack stoppers
- the detectable size — choice of method and, above all, access
- the number of load paths

A structure that is awkward to inspect is a design error, not a maintenance problem.

The three philosophies

Safe-life, fail-safe, damage tolerance

Safe-life. Assumed crack-free; retired at (demonstrated life)/(scatter factor 3–5). No inspection credit. Landing-gear structure, engine discs, rotorcraft dynamic components. Weakness: the scatter factor is a guess about variability — the F-111 wing carry-through failure in 1969, at 100 flight hours from a forging defect, is the accident that discredited it for airframes.

Fail-safe. Multiple load paths or crack-arrest features: after the complete failure of any one element the remainder carries a specified load long enough to be found. The dual-lug fitting from Lecture 1, and the tear straps that saved the Aloha aircraft.

Damage tolerance. Cracks assumed present from the first flight. Growth calculated, residual strength demonstrated, inspection programme derived. The default for transport metallic structure, and the inspection programme is part of the certified design.

WFD limits all of it

Everything here assumes *one* dominant crack in sound structure. Aloha 243 broke that assumption. Hence widespread fatigue damage in the regulations, and the Limit of Validity — a cycle count beyond which the aircraft may not fly without further substantiation, however good its inspection record.

Worked examples

Example 3.1 — Critical crack in a fuselage skin

2024-T3, hoop stress 95 MPa at limit, $K_{\mathrm{Ic}} = 37\,\mathrm{MPa}\sqrt{\mathrm{m}}$, central crack $\beta = 1$

$$a_c = \frac{1}{\pi}\left(\frac{K_{\mathrm{Ic}}}{\beta\sigma}\right)^2 = \frac{1}{\pi}\left(\frac{37}{95}\right)^2 = 48\,\mathrm{mm}$$

- So a through-crack of 97 mm tip to tip is critical — large and visible, which is the point: 2024-T3 was chosen so the critical crack is big enough to find
- Same stress in 7075-T6 ($K_{\mathrm{Ic}} = 24\,\mathrm{MPa}\sqrt{\mathrm{m}}$): $a_c = 20\,\mathrm{mm}$, less than half

Validity check

$2.5(K/\sigma_y)^2 = 2.5(37/289)^2 = 41\,\mathrm{mm} \gg 1.6\,\mathrm{mm}$ skin. The sheet is in plane stress, so the real toughness is higher and this estimate is conservative — but a thin skin properly needs an R-curve, not a single number.

Example 3.2 — Crack growth at a fastener hole

7075-T6, $\Delta\sigma = 80\,\text{MPa}$, $R = 0$, $a_i = 1.27\,\text{mm}$, $\beta = 1.12$, $C = 2.0 \times 10^{-11}$, $m = 3.5$ (m/cycle, MPa $\sqrt{\text{m}}$), $K_{\text{Ic}} = 24\,\text{MPa}\sqrt{\text{m}}$

$$a_f = \frac{1}{\pi}\left(\frac{24}{1.12 \times 80}\right)^2 = 22.8\,\text{mm}$$

$$(m-2)C\beta^m\Delta\sigma^m\pi^{m/2} = 1.5 \times 2.0{\times}10^{-11} \times 1.487 \times 4.579{\times}10^6 \times 7.413 = 1.514{\times}10^{-3}$$

$$a_i^{-0.75} - a_f^{-0.75} = 148.6 - 17.0 = 131.6 \qquad \Rightarrow \qquad N_f = \frac{2 \times 131.6}{1.514{\times}10^{-3}} = \mathbf{1.74 \times 10^5}\ \text{cycles}$$

Units — the most common error in this subject

C is published in at least four unit systems; m does not change between them, so a wrong C is out by 10^3 and still looks plausible on a log plot. **Sanity check:** aluminium at $\Delta K = 10\,\text{MPa}\sqrt{\text{m}}$ should give a few $\times 10^{-8}$ m/cycle. Here $C(10)^{3.5} = 6.3 \times 10^{-8}$ m/cycle. Passes.

Example 3.3 — Setting the interval

Same lug, eddy current with $a_{90/95} = 2\,\text{mm}$, 1 500 flights per year

$$N(1.27 \to 2\,\text{mm}) = \frac{2(148.6 - 105.7)}{1.514 \times 10^{-3}} = 5.67 \times 10^4 \text{ flights}$$

$$N(1.27 \to 22.8\,\text{mm}) = 1.74 \times 10^5 \text{ flights}$$

$$N_{\text{window}} = 1.17 \times 10^5 \text{ flights} \quad \Rightarrow \quad \text{interval} = \tfrac{1}{2} N_{\text{window}} = 5.9 \times 10^4 \ (\approx 39 \text{ years})$$

Now change one thing

The fitting is buried: general visual only, $a_{\text{det}} = 25\,\text{mm}$. But $a_c = 22.8\,\text{mm}$ — the crack is critical before it is detectable. The window is negative and no inspection interval exists.

Options: lower the stress or raise the toughness ($a_c \uparrow$), provide access ($a_{\text{det}} \downarrow$), add a second load path, or make it safe-life.

This calculation costs ten minutes and decides whether a design is maintainable. Do it while the layout can still change.

For next time

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- Shigley §§5.12 and 6.5; Dowling Ch. 8 and 11 — the best treatment at this level; Anderson Ch. 2–3
- Niu, *Airframe Structural Design*, Ch. 15 (damage tolerance)
- CS-25.571 with AMC 25.571, and FAA AC 25.571-1D for its worked treatment of inspection thresholds
- Self-study problems 1–6 in the notes (two are Python: a numerical Paris integrator with $\beta(a)$, and an interval calculation)
- Next: fatigue under combined loading — multiaxial stress, rainflow counting and Miner's rule on a real flight spectrum